NODE.JS

1.	**Create a first.js file which creation server and show the message "Hello ISTAR" as response.**

Code	var http=require('http');
	http.createServer(function(req,res){
	  res.writeHead(200,{'content-type':'text/html'});
	  res.write('Hello ISTAR!');
	  res.end();
	}).listen(8080);

2.	**Create one user-define module as say.js which contains four methods like sayCollege() which return "Hello ISTAR" and sayDepartment() which return "Hello MCA".Include this module in main.js file and use its functionality.**

Code	**<u>Say.js</u>**
	
	exports.sayCollege=function(){
	  return "Hello ISTAR";
	}
	exports.sayDepartment=function(){
	  return "Hello MCA";
	}
	
	<u>main.js</u>
	
	var http=require('http');
	var say=require('./say');
	
	http.createServer(function(req,res){
	  res.writeHead(200,{'content-type':'text/html'});
	  res.write("College:"+say.sayCollege()+"
	");
	  res.write("Department:"+say.sayDepartment());
	}).listen(8080);

3.	**Create one user define module as arith.js which contains methods like add(x,y)sub(x,y),mul(x,y),div(x,y),fact(x),sqroot(x).Includes this module in main.js and use its functionalities by passing the arguments as per prototype of method.**

Code	**<u>Airth.js</u>**
	
	exports.add=function(x,y){
	  return x+y;
	}
	exports.sub=function(x,y){
	  return x-y;
	}
	exports.mul=function(x,y){
	  return x*y;
	}
	exports.div=function(x,y){
	  return x/y;

```
}
exports.fact=function(x){
   var f=1;
   while(x>=1){
      f=f*x;
      x=x-1;
   }
   return f;
}
exports.square=function(x){
   return x*x;
}
exports.sqroot=function(x){
   return Math.sqrt(x);
}
```

main.js

```
var http=require('http');
var arith=require('./arith');

http.createServer(function(req,res){
   res.writeHead(200,{'content-type':'text/html'});

   res.write("Addition:"+arith.add(10,5)+"<br/>");

   res.write("Subtraction:"+arith.sub(10,5)+"<br/>");
   res.write("Multiplication:"+arith.mul(10,5)+"<br/>");
   res.write("Division:"+arith.div(10,5)+"<br/>");
   res.write("Square:"+arith.square(4)+"<br/>");
   res.write("SquareRoot:"+arith.sqroot(64)+"<br/>");
}).listen(8080);
```

4. **Create one user-module as string_edit.js which contains method like uc(x),lc(x),len(x).Include this module in main.js and use its functionalities by passing the arguments as per prototype of method**

Code

String_edit.js

```
exports.uc=function(x){
   return x.toUpperCase();
}
exports.lc=function(x){
   return x.toLowerCase();
}
exports.len=function(x){
   return x.length;
}
```

Main_strind.js

```
var http=require('http');
var string=require('./string_edit');
```

```
http.createServer(function(req,res){
   res.writeHead(200,{'content-type':'text/html'});
   res.write("Upper Case:"+string.uc("hello")+"<br/>");
   res.write("Lower Case:"+string.lc("HELLO")+"<br/>");
   res.write("Length of hello:"+string.len("hello"));
}).listen(8080);
```

5. **Create url_demo.js which extract the all information like host name, path name, query string from the url and display as response.**
url=http://localhost:5000/abc.htm>name=abc&name=xyz

Code

<u>url_demo.js</u>

```
var http=require('http');
var url=require('url');

http.createServer(function(req,res){
   var adr='http://localhost:8080/url_demo.js?year=2017&month=feb';
   var q=url.parse(adr,true);
   res.writeHead(200,{'content-type':'text/html'});
   res.write("Host Name:"+q.host+"<br/>");
   res.write("Path Name:"+q.pathname+"<br/>");
   res.write("Query String:"+q.search+"<br/>");
   var qu=q.query;
   res.write("year:"+qu.year+"<br/>");
   res.write("Month:"+qu.month);
}).listen(8080);
```

6. **Create fs_demo.js which will read the content from the test.js and write it in new_test.txt.**

Code

<u>Test.js</u>

```
// This is test.js
console.log("Hello from test.js!");
function greet(name) {
   return `Hello, ${name}!`;
}
console.log(greet("Node.js"));
```

<u>fs_demo.js</u>

```
const fs = require('fs');

// Read from test.js
fs.readFile('test.js', 'utf8', (err, data) => {
   if (err) {
      console.error("Error reading test.js:", err);
      return;
   }

   // Write to new_test.txt
```

```
  fs.writeFile('new_test.txt', data, (err) => {
    if (err) {
      console.error("Error writing to new_test.txt:", err);
    } else {
      console.log("Content copied from test.js to new_test.txt successfully.");
    }
  });
});
```

7.	**Create two files as mca.html and it.html. Create one department.js file which reads the request from url and display the content of the file according it. For example: if user request like http://localhost:5000/mca then it will read the mca.html file and showing the content and its response.**
Code	

Mca.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MCA Department</title>
</head>
<body>
  <h1>Welcome to the MCA Department</h1>
  <p>Here you will find all the information about the MCA programs, courses, and
more.</p>
</body>
</html>
```

It.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>IT Department</title>
</head>
<body>
  <h1>Welcome to the IT Department</h1>
  <p>Explore the IT department for technology programs, innovations, and
more!</p>
</body>
</html>
```

Department.js

```
const http = require('http');
const fs = require('fs');
const path = require('path');

const server = http.createServer((req, res) => {
```

```
  const requestedUrl = req.url;
  let filePath = '';
  if (requestedUrl === '/mca') {
    filePath = path.join(__dirname, 'mca.html'); // MCA file
  } else if (requestedUrl === '/it') {
    filePath = path.join(__dirname, 'it.html'); // IT file
  } else {
    res.statusCode = 404;
    res.end('Page not found');
    return;
  }
  fs.readFile(filePath, 'utf8', (err, data) => {
    if (err) {
      res.statusCode = 500;
      res.end('Error reading the file');
    } else {
      res.statusCode = 200;
      res.setHeader('Content-Type', 'text/html');
      res.end(data);
    }
  });
});

server.listen(5000, () => {
  console.log('Server is running at http://localhost:5000');
});
```

8.

Do as directed:

a. Create database as "data"

b. Create table as movie with the fields as mid, mname, category (Hollywood, Bollywood, Gujarati), etc.

c. Insert at least the three records for each category.

Retrieve the data category wise and store it into separate JSON file like Hollywood.json, Bollywood.json, and Gujarati.json.

Code

```
const mysql = require('mysql2');
const fs = require('fs');
const connection = mysql.createConnection({
  host: 'localhost',
  user: 'root',
  password: '', // change if needed
  multipleStatements: true
});

connection.connect(err => {
  if (err) throw err;
  console.log("Connected to MySQL");

  const createDBAndTable = `
    CREATE DATABASE IF NOT EXISTS data;
    USE data;
    CREATE TABLE IF NOT EXISTS movie (
      mid INT AUTO_INCREMENT PRIMARY KEY,
      mname VARCHAR(255),
```

```
    category VARCHAR(50)
  );
`;

connection.query(createDBAndTable, (err) => {
  if (err) throw err;

  console.log("Database and table ready.");

  const insertMovies = `
    INSERT INTO movie (mname, category) VALUES
      ('Avengers', 'Hollywood'),
      ('Inception', 'Hollywood'),
      ('Titanic', 'Hollywood'),
      ('3 Idiots', 'Bollywood'),
      ('Dangal', 'Bollywood'),
      ('Sholay', 'Bollywood'),
      ('Hellaro', 'Gujarati'),
      ('Chhello Divas', 'Gujarati'),
      ('Reva', 'Gujarati')
    ON DUPLICATE KEY UPDATE mname = VALUES(mname);
  `;

  connection.query(insertMovies, (err) => {
    if (err) throw err;
    console.log("Movies inserted.");

    // Step 4: Export to JSON
    const categories = ['Hollywood', 'Bollywood', 'Gujarati'];

    categories.forEach(category => {
      const query = `SELECT * FROM movie WHERE category = ?`;
      connection.query(query, [category], (err, results) => {
        if (err) throw err;

        fs.writeFileSync(`${category}.json`, JSON.stringify(results, null, 2));
        console.log(`${category}.json created`);
      });
    });

    // Close connection after a short delay (to allow all async writes to finish)
    setTimeout(() => connection.end(), 1000);
  });
});
});
```

9.	**Post the data as firstname and lastname using Express and display it in response page.**
Code	**<u>Index.js</u>**

```
var express = require("express");
var app = express();
app.use(express.static("public"));
```

```
// bodyParser is require for access posted data
var bodyParser = require("body-parser");
//urlencoded function
var urlencodedParser = bodyParser.urlencoded({ extended: false });

//This will call when user request index.html file.
app.get("/index.html", function (req, res) {
  res.sendFile(__dirname + "/" + "index.html");
});

//This will call only when user submit data from index.html
app.post("/index.html", urlencodedParser, function (req, res) {
  //   Prepare output in JSON format
  res_data = {
    first_name: req.body.first_name,
    last_name: req.body.last_name,
  };

  res.end(JSON.stringify(res_data));

});

var server = app.listen(8000);
console.log("http://localhost:8000/index.html");
```

index.html

```
<html>
  <body>
    <form action="http://127.0.0.1:8000/index.html" method="POST">
      First Name: <input type="text" name="first_name" /> <br />
      Last Name: <input type="text" name="last_name" />
      <input type="submit" value="Submit" />
    </form>
  </body>
</html>
```

10. Read the data from JSON file and display it in HTML template.

Code

data.html

```
<table align="center" width="50%" border="1">
  <tr>
    <td>First Name</td>
    <td>{{%FNAME%}}</td>
  </tr>
  <tr>
    <td>Last Name</td>
    <td>{{%LNAME%}}</td>
  </tr>
</table>
```

	Index.js
	var express = require("express");
	var app = express();
	app.use(express.static("public"));
	// bodyParser is require for access posted data
	var bodyParser = require("body-parser");
	//urlencoded function
	var urlencodedParser = bodyParser.urlencoded({ extended: false });
	
	var fs = require("fs");
	const htmlData = fs.readFileSync("./data.html", "utf-8");
	
	//Display data using template
	  data = htmlData.replace("{{%FNAME%}}", req.body.first_name);
	  data = data.replace("{{%LNAME%}}", req.body.last_name);
	  res.end(data);
	
	});
	
	var server = app.listen(8000);
	console.log("http://localhost:8000/index.html");